
A Fairness Beamforming Design in RSMA-ISAC System under Imperfect CSI

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A Fairness Beamforming Design in RSMA-ISAC System under Imperfect CSI

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Abstract

User fairness in the integrated sensing and communication (ISAC) system has emerged as a critical concern in green communication. Rate splitting multiple access (RSMA) offers a viable solution to enhance fairness by enabling flexible resource allocation in the ISAC system. In this paper, we considered a multi-user RSMA-ISAC system with bounded errors in the channel state information (CSI), in which the dual-functional base station (BS) can communicate with multiple users and simultaneously detect the target. A robust beamforming optimization problem is proposed aimed at maximum-minimum fairness (MMF) rates of users in the RSMA-ISAC system. To guarantee the reliability of the design, we implement robust optimization under the worst-case scenario and model the uncertain channel as a bounded ellipsoid. Due to the existence of an infinite error set, the proposed problem is non-convex and intractable to solve directly. To tackle this challenge, two solution approaches are proposed. First, the traditional alternating optimization (AO) method based on the weighted minimum mean square error (WMMSE) framework is adopted, which iteratively solves a series of convex subproblems to converge to the local optimum. Additionally, a penalty-based method utilizing semidefinite relaxation (SDR) is proposed, which rigorously handles channel uncertainty sets via the S-procedure. The numerical simulation shows that our RSMA-based scheme retains effectiveness and stability even in the presence of incomplete CSI. Moreover, the results verify that it can achieve significant performance gains compared to space division multiple access (SDMA).

Index Terms

Integrated sensing and communications (ISAC), rate-splitting multiple access (RSMA), Cramér-Rao bound (CRB), robust beamforming design.

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I. INTRODUCTION

WITH the advancement of the B5G/6G era, substantial improvements are expected in both data transmission rates and system intelligence. Sensing technology has become a critical basic service in 6G, and play an indispensable role in fields such as Internet of Vehicles (IoV), Internet of Things (IoT) and human-machine interaction. This will inevitably drive a sustained growth in the demand for limited resources. The Integrated Sensing and Communication (ISAC) system addresses this challenge by integrating communication and sensing functionalities within a single spectrum band, thereby significantly enhancing spectrum utilization efficiency [1]. Therefore, to achieve the green communication goals, ISAC is considered an advanced technological paradigm for 6G and has extensive usages in numerous scenarios.

One of the core challenges in ISAC lies in designing the appropriate waveform that can simultaneously meet multi-user communication and sensing requirements [2]–[4]. As an advanced dynamic multiple access technique, rate splitting multiple access (RSMA) has demonstrated remarkable interference management capabilities by employing linearly precoded rate-splitting and successive interference cancellation (SIC) [5]. Additionally, extensive studies have demonstrated that RSMA outperforms traditional multiple access techniques, particularly in terms of energy efficiency (EE) and power control [6], [7]. Thus, RSMA has emerged as a highly promising strategy for advancing green communications.

Combining RSMA and ISAC is widely recognized as an auspicious approach for advancing ISAC system design [8]. The research landscape of RSMA-ISAC has evolved through systematic investigations. In [9], the authors pioneered the study of fairness optimization in single target and multi-user RSMA-assisted ISAC systems. Building on this foundational work, subsequent studies extended this framework to multi-moving-target scenarios [10]. Furthermore, [11] explored the application of RSMA-assisted ISAC in low Earth orbit (LEO) satellite communications, where the primary objective was to maximize energy efficiency (EE). Meanwhile, by synergizing reconfigurable intelligent surfaces (RIS) with ISAC capabilities, [12] addresses the issue of the secure transmission challenge in the RSMA-RIS-ISAC system. However, current research has primarily assumed perfect channel state information (CSI) in the RSMA-ISAC system, while studies addressing imperfect CSI remain relatively scarce.

In practical engineering designs, imperfect CSI is far more prevalent. Specifically, the CSI of the transmitter (CSIT) will be affected by various imperfections, including the estimation errors in time division duplex (TDD) systems and the quantization error and feedback delay in frequency division duplex (FDD) systems. In Multi-User MIMO (MU-MIMO) systems, the base station (BS) relies heavily

on CSIT to design precoding schemes and steer the beam. However, imperfect CSI hinders the complete elimination of inter-user interference, leading to degraded beamforming performance. Consequently, the anti-interference capability, energy efficiency, and spectral efficiency are all compromised to a certain extent [13]. This gives rise to the critical problem of robust design against channel uncertainties. Thus, the issue of this paper is to propose a novel beamforming method that can ensure the RSMA-ISAC system maintains satisfactory performance under imperfect CSIT.

Generally, two types of channel error models are widely adopted in wireless communication: statistical error models and ellipsoidal bounded uncertainty model. Both models are essential for characterizing the uncertainty of CSI in wireless communication, each featuring distinct applicability scenarios suited to different system requirements. The former regards the channel estimation error as a random variable with a specific probability distribution, making it suitable for relatively stable channel environments in which statistical characteristics can be reliably approximated. In contrast, the latter constrains the error set within an ellipsoidal boundary, thereby enhancing robustness to unmodeled dynamics. Thus, it is particularly appropriate for complex and time-varying channel scenarios in which accurate modeling of channel error statistics is challenging [5]. In the context of RSMA-ISAC systems, relevant research has explored these error models to address practical performance challenges. The authors investigated the performance of RSMA-ISAC under the statistical error model and designed a sensing subsystem based on radar beam matching in [14]. Zhang *et al.* [15] proposed a robust secure beamforming design for RSMA-RIS-ISAC systems that explicitly incorporates the bounded error model to mitigate CSI uncertainty. By additionally considering the secrecy outage probability (SOP), Peng *et al.* took the signal-to-interference-plus-noise ratio (SINR) of the radar as the optimization objective [16].

The max-min fairness (MMF) problem can be characterized by the maximum-minimum user rate, which is a key metric for evaluating the balance and equitable distribution of resources among users [17]. To our knowledge, the MMF problem under the bounded error model has not been previously addressed, which motivates our research. The performance of RSMA-ISAC systems under the bounded uncertainty model is explored to address the user fairness challenge. In this paper, we first investigate the MMF rate problem in the RSMA-ISAC system under ellipsoidal bounded uncertain channels and propose two approaches. The Cramér-Rao bound (CRB) of unbiased radar estimation is adopted as a performance requirement for the sensing subsystem. The first approach considers the robust optimization problem in the worst case scenario, where precoders are designed to guarantee a certain level of performance for all possible channels

within the corresponding uncertainty regions. The perturbations of the worst case are eliminated by the S-procedure in the second method, and the system stability across the entire uncertainty range is guaranteed. Consequently, the originally intractable problem induced by imperfect channel state information at the transmitter (CSIT) is transformed into a computationally tractable deterministic problem.

The main contributions of this paper are as follows:

- The beamforming design problem for RSMA-ISAC systems under imperfect CIST is first investigated to meet user communication requirements and target sensing tasks simultaneously. To achieve robust optimization under the worst-case scenario, an ellipsoidal bounded uncertainty model is adopted to characterize the magnitude of channel estimation errors. An optimization problem is formulated with the MMF rate as the objective function, subject to constraints on transmit power, estimation CRB, and channel uncertainty.
- To address this problem, Alternating optimization (AO) algorithm based on the weighted minimum mean square error (WMMSE) framework is employed. First, the original rate-related problem is transformed into an equivalent WMMSE form to eliminate non-convexity in the rate expressions. And then, the problem is reformulated via the Schur complement theorem and its generalized form to handle the channel uncertainty. At last, the solution is obtained via iterative AO. Numerical simulation results demonstrate that the RSMA-ISAC scheme under imperfect CSI can achieve superior user fairness and enhances overall system performance by using AO algorithm.
- Additionally, to reduce the complexity the double-layer penalty-based semidefinite relaxation (SDR) algorithm is proposed in this paper. The successive convex approximation (SCA) technique and S-procedure are leveraged to convert the original non-convex problem into a computationally tractable convex form. To ensure solution feasibility, the SCA serves as the inner loop while a rank-one penalty term operates as the outer loop in this structure. Numerical simulation shows that the S-procedure method outperforms the traditional AO-WMMSE scheme in complexity and system performance. Furthermore, this algorithm offers more effective solution in practical deployment scenarios involving channel uncertainty.

The rest of this paper is organized as follows. The system model is describes briefly in Section II. Section III formulates the MMF optimization problem and elaborates on the relevant performance metrics involved in the problem formulation. Section IV and Section V present in detail the two proposed algorithms. Finally, Section VI provides the simulation results and Section VII concludes this paper.

Notations: Matrices, column vectors, and scalars are denoted by bold uppercase, bold lowercase, and regular (non-bold) letters, respectively. $\mathbf{I}$ and $\mathbf{0}$ donate the identity matrix and the zero matrix. $(\dot{x})$ represents the derivative of x . $\mathbb{C}^{M \times N}$ represents the space of $M \times N$ complex matrix. $\text{diag}(\mathbf{x})$ denotes a matrix with the entries of vector $\mathbf{x}$ placed along the diagonal. $\text{Tr}(\cdot)$, $(\cdot)^H$, $(\cdot)^T$, $(\cdot)^*$, $(\cdot)^{-1}$, and $\text{rank}(\cdot)$ represent trace, conjugate-transpose, transpose, conjugate, inverse and rank, respectively. $|\cdot|$, $\|\cdot\|$ and $\mathbb{E}(\cdot)$ are utilized to denote absolute value, Euclidean norm and statistical expectation. $\mathbf{A} \succcurlyeq 0$ means that the matrix $\mathbf{A}$ is positive semidefinite.

II. SYSTEM MODEL

In this paper, a downlink RSMA-ISAC system is considered in Fig. 1, where a dual-function base station (BS) equipped with N_t transmit antennas and N_r receive antennas serves K single-antenna users, indexed by $\mathcal{K} = \{1, 2, \dots, K\}$, and detects a radar target. To ensure the feasibility of the beamforming design and the accuracy of the radar reception information, assume $K \leq N_t < N_r$.

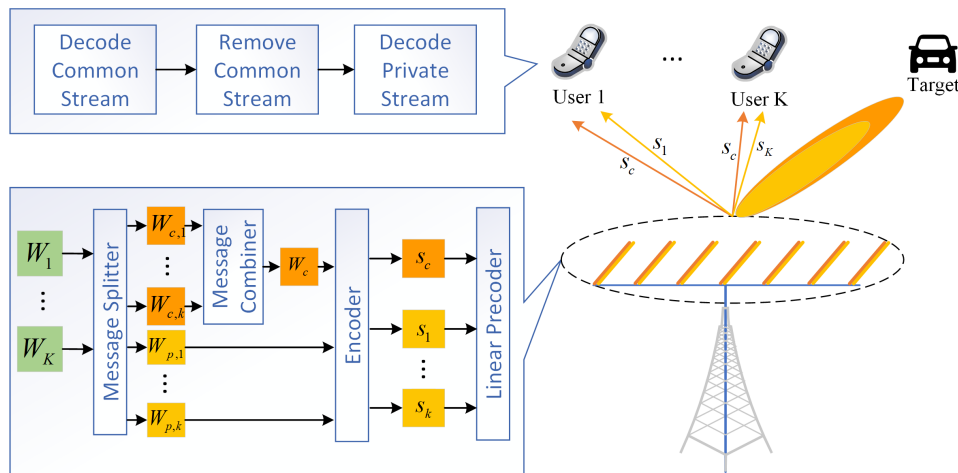


Fig. 1: The architecture of the RSMA-assisted ISAC system.

We use 1-layer RSMA [5] to split the message W_k of the user k into the common message $W_{c,k}$ and the private message $W_{p,k}$, i.e., $W_k = \{W_{c,k}, W_{p,k}\}$. All common messages $W_{c,k}$ are combined into one common message $W_c = \{W_{c,1}, W_{c,2}, \dots, W_{c,K}\}$. Then, the common message W_c and the private message W_k of each user are encoded into the common stream s_c and the private stream $s_{p,k}$, respectively. The transmitted signal $\mathbf{s}$ can be expressed as $\mathbf{s}(l) = \{s_c(l), s_1(l), s_2(l), \dots, s_k(l)\} \in \mathbb{C}^{(K+1) \times 1}$, where $l = \{1, \dots, L\} \in \mathcal{L}$. From the communication perspective, l represents the time slot, whereas from the radar perspective, l denotes the fast-time snapshot. Focusing on the l time slot, all data streams are transmitted through the beamforming precoder $\mathbf{P} = [\mathbf{p}_c, \mathbf{p}_1, \dots, \mathbf{p}_k] \in \mathbb{C}^{N_t \times (K+1)}$, which $\mathbf{p}_c \in \mathbb{C}^{N_t \times 1}$ is the beamformer of the

common stream and $\mathbf{p}_k \in \mathbb{C}^{N_t \times 1}$ the beamformer of the private stream. Thus, the baseband transmit signal can be entirely represented as:

$$\mathbf{x}(l) = \mathbf{p}_c s_c(l) + \sum_{i=1}^K \mathbf{p}_i s_i(l). \quad (1)$$

First, the data streams are assumed to be independent of each other, so that $\mathbf{s}(l)$ should be satisfied by $\mathbf{s}(l)\mathbf{s}(l)^H = \mathbf{I}_{K+1}$. Then, the signal covariance matrix can be derived as $\mathbf{R}_x = \frac{1}{L} \sum_{l \in \mathcal{L}} \mathbf{x}(l)\mathbf{x}(l)^H = \mathbf{P}\mathbf{P}^H$. And the power constraint of the transmitted signal with total power P_t can be expressed as $\|\mathbf{p}_c\|^2 + \sum_{i=1}^K \|\mathbf{p}_i\|^2 \leq P_t$. The received signal of the user k is $y_k(l) = \mathbf{h}_k^H \mathbf{x}(l) + n_k(l)$. Thus, we can get:

$$y_k(l) = \mathbf{h}_k^H \mathbf{p}_c s_c(l) + \sum_{i=1}^K \mathbf{p}_i s_i(l) + n_k(l), \forall k \in \mathcal{K} \quad (2)$$

where $\mathbf{h}_k = g_k \Lambda_k^{\frac{1}{2}}, \forall k \in \mathcal{K}$ represents the channel vector between the BS and the user k , $g_k \in \mathbb{C}^{N_t \times 1}$ and $\Lambda_k^{\frac{1}{2}}$ denote the large and small scale fading, respectively. Suppose $n_k(l)$ denotes the received noise which is the complex Gaussian noise with zero mean and variance σ_n^2 .

If given the target response matrix $\mathbf{G}(\theta)$, the reflected echo signal $y_r(l)$ at the receiver of the BS is given by:

$$y_r(l) = \mathbf{G}(\theta) \mathbf{x}(l) + n_r(l), \quad (3)$$

where $n_r(l)$ denotes an additive white Gaussian noise(AWGN) matrix with variance σ_R^2 for each entry.

Suppose α represents the reflection coefficient, which need incorporate both the round-trip path loss and the radar cross-section (RCS) of the target. If θ denotes the azimuth angle of the target, while $\mathbf{a}(\theta) = [1, e^{j\pi \sin(\theta)}, \dots, e^{j\pi(N_t-1)\sin(\theta)}]^H \in \mathbb{C}^{N_t \times 1}$ and $\mathbf{b}(\theta) = [1, e^{j\pi \sin(\theta)}, \dots, e^{j\pi(N_r-1)\sin(\theta)}]^H \in \mathbb{C}^{N_r \times 1}$ are the steering vectors of the transmit and receive antenna arrays. When define $\mathbf{A}(\theta) \triangleq \mathbf{b}(\theta)\mathbf{a}^H(\theta)$ [15], the target response matrix $\mathbf{G}(\theta)$ can be expressed as :

$$\mathbf{G}(\theta) = \alpha \mathbf{A}(\theta). \quad (4)$$

III. PERFORMANCE METRICS AND PROBLEM FORMULATION

A. Communication metric

To reflect the fairness of users, the maximum and minimum user rates is used as the fairness index [17]. At the receiver side, the user first decodes the common stream. All private streams are treated as

interference. Thus, the common stream is removed by using successive interference cancellation (SIC). Then, the users decode their private streams by treating the other private streams as interference. The common stream is decoded first, and then the private data. Based on (2), the signal-to-interference-plus-noise ratio (SINR) in decoding $s_c(l)$ and $s_k(l)$ of the user k can be expressed as:

$$\gamma_{c,k}(\mathbf{h}_k, \mathbf{P}) = \frac{|\mathbf{h}_k^H \mathbf{p}_c|^2}{\sum_{j=1}^K |\mathbf{h}_k^H \mathbf{p}_j|^2 + \sigma_n^2}, \forall k \in \mathcal{K} \quad (5)$$

$$\gamma_k(\mathbf{h}_k, \mathbf{P}) = \frac{|\mathbf{h}_k^H \mathbf{p}_k|^2}{\sum_{j=1, j \neq k}^K |\mathbf{h}_k^H \mathbf{p}_j|^2 + \sigma_n^2}, \forall k \in \mathcal{K} \quad (6)$$

Then, the achievable rate corresponding to the SINR can be derived as follows:

$$R_{c,k}(\mathbf{h}_k, \mathbf{P}) = \log_2(1 + \gamma_{c,k}), \quad (7)$$

$$R_k(\mathbf{h}_k, \mathbf{P}) = \log_2(1 + \gamma_k). \quad (8)$$

It is worth noting that all users need to decode the common stream successfully. Thus, similar to the rate must be satisfied $R_c = \min\{R_{c,k}(\mathbf{h}_k, \mathbf{P})\}, \forall k \in \mathcal{K}$ [12]. Since R_c is shared by all users, the assigned portion of the common rate for each user can be represented by $C_k, k \in \mathcal{K}$. Then the common rate for each user can be expressed as (9):

$$R_c = \sum_{k=1}^K C_k. \quad (9)$$

The achievable rate of the user k includes the common rate C_k , as well as the private rate $R_k(\mathbf{h}_k, \mathbf{P})$. Therefore, the achievable rate of the user k can be expressed as $C_k + R_k(\mathbf{h}_k, \mathbf{P})$. Our goal is to maximize the minimum achievable rate among all users, ensuring the rate of each user is the same and thereby maintaining fairness.

B. Sensing metric

Suppose that the detected target adheres to the point target model, which simplifies the analysis by treating the target as a single point with negligible spatial extent [16]. To evaluate and optimize the performance of the radar system, we adopt CRB as the radar sensing metric. The CRB provides a

theoretical lower bound on the variance of the unbiased estimators for target parameters, such as range, velocity, and angle, which serves as a fundamental benchmark for assessing the accuracy and resolution of the radar system [18]. By minimizing the CRB, the precise estimation capability of the system for the target characteristics can be enhanced, the robust and reliable sensing performance will be ensured.

The CRB is fundamentally defined as the inverse of the Fisher Information Matrix (FIM). The FIM quantifies the amount of information the observed data provides on the unknown parameters. Specifically, the CRB is derived as $\text{CRB} = \mathbf{J}^{-1}$, where $\mathbf{J}$ represents the FIM. Let $\Xi = [\theta \ \alpha]^T$ represent the vector of unknown parameters. For the parameter vector Ξ , the FIM can be expressed as:

$$\mathbf{J}[\Xi] = \mathbb{E}\left[\left(\frac{\partial \ln p(\mathbf{y}|\Xi)}{\partial \Xi}\right)\left(\frac{\partial \ln p(\mathbf{y}|\Xi)}{\partial \Xi}\right)^H\right], \quad (10)$$

where $\ln p(\mathbf{y}|\Xi)$ is the likelihood function of the observed data $\mathbf{y}$ given the parameter vector Ξ . According to the assumptions stated in the previous section, the measurement vectors are independent complex Gaussian vectors with $y_r(l) \sim \mathcal{N}(\alpha \mathbf{A}(\theta) \mathbf{x}(l), \sigma_R^2 \mathbf{I}_{N_r})$. To simplify the expression of $\mathbf{J}$, according to the model in (3), $\mathbf{J}$ can be further expressed as:

$$\mathbf{J}[\Xi]_{m,n} = \frac{2}{\sigma_R^2} \mathcal{R}\left\{\left(\frac{\partial v(l)}{\partial \Xi_m}\right)^H \left(\frac{\partial v(l)}{\partial \Xi_n}\right)\right\}, \quad m, n = 1, 2 \quad (11)$$

where $v(l) \triangleq \alpha \mathbf{A}(\theta) \mathbf{x}(l)$. Therefore, the FIM for estimating the parameters can be partitioned as:

$$\mathbf{J}[\Xi] = \begin{bmatrix} J_{\theta\theta} & J_{\theta\alpha} \\ J_{\theta\alpha}^T & J_{\alpha\alpha} \end{bmatrix}, \quad (12)$$

and the CRB for DOA estimate can be expressed as:

$$\text{CRB}(\theta) = [J_{\theta\theta} - J_{\theta\alpha} J_{\alpha\alpha}^{-1} J_{\theta\alpha}^T]^{-1}, \quad (13)$$

where $J_{\theta\theta}$, $J_{\theta\alpha}$ and $J_{\alpha\alpha}$ is given by:

$$J_{\theta\theta} = \frac{2L\alpha^2}{\sigma_R^2} \mathcal{R}\{Tr(\dot{\mathbf{A}}(\theta) \mathbf{R}_x \dot{\mathbf{A}}^H(\theta))\}, \quad (14)$$

$$J_{\theta\alpha} = \frac{2L}{\sigma_R^2} \mathcal{R}\{\alpha^* Tr(\mathbf{A}(\theta) \mathbf{R}_x \dot{\mathbf{A}}^H(\theta))(1, j)\}, \quad (15)$$

$$J_{\alpha\alpha} = \frac{2L}{\sigma_R^2} \mathcal{R}\{(1, j)^H (1, j) Tr(\mathbf{A}(\theta) \mathbf{R}_x \mathbf{A}^H(\theta))\}. \quad (16)$$

Substituting these parameters into (13), the CRB for the DOA angle estimation can be derived in (17).

Here, the SNR of the received radar echo is defined as $\text{SNR}_{\text{radar}} \triangleq \frac{P_t L |\alpha|^2}{\sigma_R^2}$ [16]. Additionally, $\dot{\mathbf{A}}(\theta)$ represents the derivative of $\mathbf{A}(\theta)$ with respect of θ , and can be expressed as $\dot{\mathbf{A}}(\theta) = \frac{\partial \mathbf{A}(\theta)}{\partial \theta}$. The value of $\text{CRB}(\theta)$ depends on θ and $\mathbf{R}_x$. Assuming that θ remains constant throughout the optimization process, minimizing $\text{CRB}(\theta)$ can be interpreted as a beamforming design optimized for a specific direction of interest, where θ represents the potential angle at which a target may be located [14]. This approach is known as beam prediction, which is particularly common in target tracking scenarios within vehicular networks. The radar system aims to focus its beam toward an estimated or predicted direction to track moving targets accurately.

$$\text{CRB}(\theta) = \frac{\text{Tr}(\mathbf{A}(\theta) \mathbf{R}_x \mathbf{A}^H(\theta))}{2\text{SNR}(\text{Tr}(\dot{\mathbf{A}}(\theta) \mathbf{R}_x \dot{\mathbf{A}}^H(\theta)) \text{Tr}(\mathbf{A}(\theta) \mathbf{R}_x \mathbf{A}^H(\theta)) - |\text{Tr}(\mathbf{A}(\theta) \mathbf{R}_x \dot{\mathbf{A}}^H(\theta))|^2)}. \quad (17)$$

C. Channel state information

Due to uncertainties such as channel estimation errors, feedback delay, and quantization errors in actual communication systems, it is difficult to obtain the perfect CSI. What's more, CSIR can be acquired with near-perfect accuracy due to the UE's capacity for real-time channel estimation via downlink pilots, coupled with its relatively simpler radio frequency (RF) hardware exhibiting lower phase noise and fewer hardware impairments. Conversely, achieving perfect CSIT remains challenging: in FDD systems, reliance on user feedback introduces quantization errors, latency, and compression artifacts, while in TDD systems, despite channel reciprocity, hardware non-idealities cause discrepancies between uplink-estimated and actual downlink channels, further compounded by pilot contamination and time-varying channel effects, ultimately resulting in imperfect CSIT. Therefore, a channel model is assumed where the CSIT is imperfect, while the CSIR is considered perfect [16]. A typical imperfect CSIT model can be described as:

$$\mathbf{h}_k = \hat{\mathbf{h}}_k + \tilde{\mathbf{h}}_k, \quad (18)$$

where $\hat{\mathbf{h}}_k$ is the estimated channel, and $\tilde{\mathbf{h}}_k$ is the channel estimation error. In channel uncertainty modeling, it is typically necessary to select an appropriate uncertainty model based on different network scenarios. To obtain more robust optimization design results, we adopt the ellipsoidal bounded uncertainty channel for the worst-case scenario [19]. Under this model, the channel estimation errors $\tilde{\mathbf{h}}_k$ is confined within an ellipsoid of radius $\varepsilon_{e,k}$. Therefore, the uncertain channel set can be defined as [14]:

$$\mathbb{H}_k \triangleq \{\mathbf{h}_k | \mathbf{h}_k = \hat{\mathbf{h}}_k + \tilde{\mathbf{h}}_k, \|\tilde{\mathbf{h}}_k\|^2 \leq \varepsilon_{e,k}^2\}. \quad (19)$$

To mitigate the impact of such uncertainties, a robust optimization approach is developed for the worst-case scenario. The bounded achievable rates can be derived from the bounded uncertainties to ensure the reliability of the system by using this approach. Additionally, the second method is proposed, in which the S-procedure is employed to address these uncertainties directly. The S-procedure enables the reformulation of quadratic uncertainty constraints into linear matrix inequality (LMI), rendering the problem computationally tractable.

D. Max-min fairness rate optimization problem

The primary objective of this paper is to maximum fairness rate among users while maintaining the accuracy of radar echo detection. Therefore, the optimization problem can be formulated as:

$$\max_{\mathbf{P}, \mathbf{c}} \min (C_k + R_k(\mathbf{h}_k, \mathbf{P})) \quad (20)$$

$$\text{s. t. } C_k \geq 0, \quad \forall k \in \mathcal{K} \quad (20a)$$

$$\sum_{i=1}^K C_k \leq R_{c,k}(\mathbf{h}_k, \mathbf{P}), \quad \forall k \in \mathcal{K} \quad (20b)$$

$$\text{diag}(\mathbf{P}\mathbf{P}^H) = \frac{P_t \mathbf{1}_{N_t \times 1}}{N_t} \quad (20c)$$

$$\text{CRB}(\theta) \leq \rho \quad (20d)$$

$$\mathbf{h}_k \in \mathbb{H}_k \quad (20e)$$

where $\mathbf{c} = [C_1, C_2, \dots, C_K]$. Here, the communication constraints under the RSMA-ISAC framework are shown in (20a) and (20b). (20a) can ensure the minimum value of the common rate, while (20b) guarantees that all users can successfully decode the final common messages. Unlike traditional communication optimization problems, (20c) and (20d) are related to the radar system. The constant modulus constraint of the sensing system is shown in (20c). (20d) represents the CRB constraint for the radar system. The smaller the value of ρ , the more accurate the parameter estimation. The channel uncertainty set is represented by (20e).

This non-convex optimization problem involves uncertainties, which make it challenging to solve directly. In this paper, two solutions are proposed to address this issue. The rate optimization problem is transformed into a minimum mean square error (MMSE) problem and solved using the AO method in the first approach. SDR combined with the S-procedure is employed to derive an efficient solution in the second approach.

IV. WORST-CASE OPTIMIZATION APPROACH

Bounded for the imperfect CSIT leads to bounded uncertainty about the achievable rates of users. To ensure successful decoding for all possible channels within the uncertainty region during resource allocation, robust optimization can be performed based on the achievable rate in the worst-case scenario. However, directly determining the worst-case rate is a challenge. The non-convex nature of the rate function makes the representation and solution further complicated. To address this, the solution of solving the maximum mean square error (MSE) is transformed into the problem of the minimum achievable rate by employing the WMMSE framework.

A. WMMSE framework

The key characteristic of the optimization problem (20) is the complicated non-convex. To address this challenge, as the approach employed in [18], the original problem can be reformulated into a WMMSE framework and then solved. Thus, the rate expression is first derived through conservative estimation for the channel uncertainties in the worst-case scenario. Subsequently, the transformed problem is solved using the AO algorithm.

For the user k , the common stream s_c is decoded first. The estimate of s_c is given as $\hat{s}_{c,k} = g_{c,k}y_k$, where $g_{c,k}$ is the equalizer. After the common stream is successfully decoded and removed via SIC, the estimated private stream s_k is obtained applying g_k such that $\hat{s}_k = g_k(y_k - \mathbf{h}_k^H \mathbf{p}_c s_c)$. The MSE of the common stream and the private stream can be defined as $\varepsilon_{c,k} \triangleq \mathbb{E}\{|\hat{s}_{c,k} - s_c|^2\}$ and $\varepsilon_k \triangleq \mathbb{E}\{|\hat{s}_k - s_k|^2\}$, respectively. Thus, they can be written as:

$$\varepsilon_{c,k} = |g_{c,k}|^2(|\mathbf{h}_k^H \mathbf{p}_c|^2 + \sum_{j=1}^K |\mathbf{h}_k^H \mathbf{p}_j|^2 + \sigma_n^2) - 2\mathcal{R}\{g_{c,k} \mathbf{h}_k^H \mathbf{p}_c\} + 1, \quad (21)$$

$$\varepsilon_k = |g_k|^2(\sum_{j=1}^K |\mathbf{h}_k^H \mathbf{p}_j|^2 + \sigma_n^2) - 2\mathcal{R}\{g_k \mathbf{h}_k^H \mathbf{p}_k\} + 1. \quad (22)$$

The MMSE of the equalizers can be obtained from $\frac{\partial \varepsilon_{c,k}}{\partial g_{c,k}} = 0$ and $\frac{\partial \varepsilon_k}{\partial g_k} = 0$, which can be expressed as:

$$g_{c,k}^{MMSE} = \frac{\mathbf{p}_c^H \mathbf{h}_k}{|\mathbf{h}_k^H \mathbf{p}_c|^2 + \sum_{j=1}^K |\mathbf{h}_k^H \mathbf{p}_j|^2 + \sigma_n^2}, \quad (23)$$

$$g_k^{MMSE} = \frac{\mathbf{p}_k^H \mathbf{h}_k}{\sum_{j=1}^K |\mathbf{h}_k^H \mathbf{p}_j|^2 + \sigma_n^2}. \quad (24)$$

By substituting (23) and (24) into (21) and (22) respectively, the MMSE is defined as $\min_{g_{c,k}} \varepsilon_{c,k}$. Then, we can get:

$$\varepsilon_{c,k}^{MMSE} = \frac{\sum_{j=1}^K |\mathbf{h}_k^H \mathbf{p}_j|^2 + \sigma_n^2}{|\mathbf{h}_k^H \mathbf{p}_c|^2 + \sum_{j=1}^K |\mathbf{h}_k^H \mathbf{p}_j|^2 + \sigma_n^2}, \quad (25)$$

$$\varepsilon_k^{MMSE} = \frac{\sum_{j=1, j \neq k}^K |\mathbf{h}_k^H \mathbf{p}_j|^2 + \sigma_n^2}{\sum_{j=1}^K |\mathbf{h}_k^H \mathbf{p}_j|^2 + \sigma_n^2}. \quad (26)$$

The relationship between MMSE and SINR can be readily derived from (25) and (26). Specifically, $\gamma_{c,k}(\mathbf{h}_k, \mathbf{P}) = (\varepsilon_{c,k}^{MMSE})^{-1} - 1$ and $\gamma_k(\mathbf{h}_k, \mathbf{P}) = (\varepsilon_k^{MMSE})^{-1} - 1$. Therefore, the rate function can also be expressed in terms of MMSE such that $R_{c,k}(\mathbf{h}_k, \mathbf{P}) = \log_2((\varepsilon_{c,k}^{MMSE})^{-1})$ and $R_k(\mathbf{h}_k, \mathbf{P}) = \log_2((\varepsilon_k^{MMSE})^{-1})$. As reported in [22], by introducing weights $(w_{c,k}, w_k)$, the common and private weighted MSE (WMSE) are given by:

$$\xi_{c,k} \triangleq w_{c,k} \varepsilon_{c,k} - \log_2(w_{c,k}), \quad (27)$$

$$\xi_k \triangleq w_k \varepsilon_k - \log_2(w_k). \quad (28)$$

By substituting the optimal receivers $g_{c,k}^{MMSE}$ and g_k^{MMSE} , the optimal weights of WMMSE $w_{c,k} = (\varepsilon_{c,k}^{MMSE})^{-1}$ and $w_k = (\varepsilon_k^{MMSE})^{-1}$ are derived by solving the first-order conditions $\frac{\partial \xi_{c,k}(g_{c,k}^{MMSE})}{\partial w_{c,k}} = 0$ and $\frac{\partial \xi_k(g_k^{MMSE})}{\partial w_k} = 0$. This yields the minimal WMMSE expression:

$$\xi_{c,k}^{MMSE} \triangleq \min_{w_{c,k}, g_{c,k}} \xi_{c,k} = 1 + \log_2(\varepsilon_{c,k}^{MMSE}) = 1 - R_{c,k}(\mathbf{h}_k, \mathbf{P}), \quad (29)$$

$$\xi_k^{MMSE} \triangleq \min_{w_k, g_k} \xi_k = 1 + \log_2(\varepsilon_k^{MMSE}) = 1 - R_k(\mathbf{h}_k, \mathbf{P}). \quad (30)$$

Considering the channel estimation errors, the robust design is adopted based on the worst-case criterion.

Consequently, the lower bounds of the common stream rates and the private stream rates can be expressed as:

$$R_{c,k}^{lb}(\mathbf{h}_k, \mathbf{P}) \triangleq \min_{\tilde{\mathbf{h}}_k} R_{c,k}(\mathbf{h}_k, \mathbf{P}) = 1 - \max_{\tilde{\mathbf{h}}_k} \min_{w_{c,k}, g_{c,k}} \xi_{c,k}, \quad (31)$$

$$R_k^{lb}(\mathbf{h}_k, \mathbf{P}) \triangleq \min_{\tilde{\mathbf{h}}_k} R_k(\mathbf{h}_k, \mathbf{P}) = 1 - \max_{\tilde{\mathbf{h}}_k} \min_{w_k, g_k} \xi_k. \quad (32)$$

By invoking the inequality [23]:

$$\max_{\tilde{\mathbf{h}}_k} \min_{w_n, g_n} \xi_n \leq \min_{w_n, g_n} \max_{\tilde{\mathbf{h}}_k} \xi_n \quad (33)$$

The achievable rates in the worst-case scenario can be expressed in the WMMSE form. The specific formulation is given by:

$$\tilde{R}_{c,k}(\mathbf{h}_k, \mathbf{P}) = 1 - \min_{w_{c,k}, g_{c,k}} \max_{\tilde{\mathbf{h}}_k} \xi_{c,k} \leq R_{c,k}^{lb}(\mathbf{h}_k, \mathbf{P}), \quad (34)$$

$$\tilde{R}_k(\mathbf{h}_k, \mathbf{P}) = 1 - \min_{w_k, g_k} \max_{\tilde{\mathbf{h}}_k} \xi_k \leq R_k^{lb}(\mathbf{h}_k, \mathbf{P}). \quad (35)$$

By substituting (34) and (35) into (20), an equivalent WMMSE formulation with a block-convex structure is obtained. This structure can be exploited via the AO algorithm to derive the solution. Consequently, the original problem becomes tractable once the uncertainty induced by the channel estimation is properly handled.

B. MMSE under the uncertainty

Based on the rate-MMSE relationship established earlier, the rate optimization problem is first transformed into an equivalent MMSE formulation to eliminate the impact of channel uncertainty [24]. By defining slack variables $\varpi_{c,k} = (g_{c,k} \hat{\mathbf{h}}_k^H \mathbf{P} - \mathbf{q}_c^H)^H$, $\psi_{c,k} = g_{c,k}^H \mathbf{P}^H$, and $\mathbf{q}_c^H = [1, 0, \dots, 0] \in \mathbb{C}^{(K+1) \times 1}$, (21) is reformulated as:

$$\varepsilon_{c,k} = |g_{c,k}|^2 \sigma_n^2 + \|\varpi_{c,k} + \psi_{c,k} \tilde{\mathbf{h}}_k\|^2. \quad (36)$$

By introducing $\varpi_k = (g_k \hat{\mathbf{h}}_k^H \mathbf{P}_{(k)} - \mathbf{q}_k^H)^H$ and $\psi_k = g_k^H \mathbf{P}_{(k)}^H$, the private MSE can also be obtained:

$$\varepsilon_k = |g_k|^2 \sigma_n^2 + \|\varpi_k + \psi_k \tilde{\mathbf{h}}_k\|^2, \quad (37)$$

where $\mathbf{P}_{(k)} = [\mathbf{p}_1, \dots, \mathbf{p}_k] \in \mathbb{C}^{N_t \times K}$ denote the precoding matrix excluding the common precoder and $\mathbf{q}_k = [0, \dots, \underset{k\text{-th}}{1}, \dots, 0] \in \mathbb{C}^{K \times 1}$ represent the selection vector whose k -th element is one, with all other entries being zeros.

Thus, the MMSE problem under the uncertainty is obtained, specifically. It can be given by:

$$\begin{aligned} \min_{g_{c,k}} \quad & \max_{\tilde{\mathbf{h}}_k} (|g_{c,k}|^2 \sigma_n^2 + \|\varpi_{c,k} + \psi_{c,k} \tilde{\mathbf{h}}_k\|^2) \\ \text{s. t.} \quad & (20e) \end{aligned} \quad (38)$$

$$\begin{aligned} \min_{g_k} \quad & \max_{\tilde{\mathbf{h}}_k} (|g_k|^2 \sigma_n^2 + \|\varpi_k + \psi_k \tilde{\mathbf{h}}_k\|^2) \\ \text{s. t.} \quad & (20e) \end{aligned} \quad (39)$$

By solving (38) and (39), the MMSE of the optimal receiver can be obtained, $g_{c,k}$ and g_k , which represent the rate within the WMMSE framework under the worst-case scenario. Taking (38) as an example, the transformation process is demonstrated. Through the introduction of the slack variable $\zeta_{c,k}$, satisfying $\zeta_{c,k} \geq \|\varpi_{c,k} + \psi_{c,k} \tilde{\mathbf{h}}_k\|^2$, (38) can be reformulated as follows:

$$\min_{g_{c,k}} \quad \max_{\tilde{\mathbf{h}}_k} (|g_{c,k}|^2 \sigma_n^2 + \zeta_{c,k}) \quad (40)$$

$$\begin{aligned} \text{s. t.} \quad & \zeta_{c,k} \geq \|\varpi_{c,k} + \psi_{c,k} \tilde{\mathbf{h}}_k\|^2 \\ & (20e) \end{aligned} \quad (40a)$$

Equation (39) can be transformed into LMI forms for (39a) and (39b) following the methodology proposed in [26]. Initially, by using Schur complement, (40a) can be transformed as follows:

$$\begin{bmatrix} \zeta_{c,k} & (\varpi_{c,k} + \psi_{c,k} \tilde{\mathbf{h}}_k)^H \\ \varpi_{c,k} + \psi_{c,k} \tilde{\mathbf{h}}_k & \mathbf{I} \end{bmatrix} \succcurlyeq 0. \quad (41)$$

And (41) can be transformed into LMI form by applying **Lemma 1** [25].

Lemma 1: Given matrix $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{A}$ with $\mathbf{A} = \mathbf{A}^H$. The relation is valid:

$$\mathbf{A} \geq \mathbf{B}^H \mathbf{D} \mathbf{C} + (\mathbf{B}^H \mathbf{D} \mathbf{C})^H, \forall \mathbf{D} : \|\mathbf{D}\| \leq \epsilon$$

if and only if there exists $\lambda \geq 0$ such that

$$\begin{bmatrix} \mathbf{A} - \lambda \mathbf{C}^H \mathbf{C} & -\epsilon \mathbf{B}^H \\ -\epsilon \mathbf{B} & \lambda \mathbf{I} \end{bmatrix} \succcurlyeq 0.$$

We can directly apply **Lemma 1** with $\mathbf{A} = \begin{bmatrix} \zeta_{c,k} & \varpi_{c,k}^H \\ \varpi_{c,k} & \mathbf{I} \end{bmatrix}$, $\mathbf{B} = \begin{bmatrix} \mathbf{0} & \psi_{c,k}^H \end{bmatrix}$, $\mathbf{C} = -\begin{bmatrix} 1 & \mathbf{0} \end{bmatrix}$, $\mathbf{D} = \tilde{\mathbf{h}}_k$, $\epsilon = \epsilon_{e,k}$. Thus, by giving fixed precoders $\mathbf{p}$, (39) can be reformulated as a set of semidefinite programming (SDP) problems:

$$\min_{g_{c,k}, \zeta_{c,k}, \Delta_{c,k}} |g_{c,k}|^2 \sigma_n^2 + \zeta_{c,k} \quad (42)$$

$$\text{s.t.} \quad \begin{bmatrix} \zeta_{c,k} - \Delta_{c,k} & \varpi_{c,k}^H & \mathbf{0} \\ \varpi_{c,k} & \mathbf{I} & -\epsilon_{e,k} \psi_{c,k} \\ \mathbf{0} & -\epsilon_{e,k} \psi_{c,k}^H & \Delta_{c,k} \mathbf{I} \end{bmatrix} \succcurlyeq 0 \quad (42a)$$

where $\Delta_{c,k} \geq 0$ are the slack variables.

This problem can be efficiently solved by using the interior-point method, and the optimal common receiver $\hat{g}_{c,k}$ can be obtained. Likewise, by defining the slack variable ζ_k , the private MMSE can be obtained as follows:

$$\min_{g_k, \zeta_k, \Delta_k} |g_k|^2 \sigma_n^2 + \zeta_k \quad (43)$$

$$\text{s.t.} \quad \begin{bmatrix} \zeta_k - \Delta_k & \varpi_k^H & \mathbf{0} \\ \varpi_k & \mathbf{I} & -\epsilon_{e,k} \psi_k \\ \mathbf{0} & -\epsilon_{e,k} \psi_k^H & \Delta_k \mathbf{I} \end{bmatrix} \succcurlyeq 0 \quad (43a)$$

where $\Delta_k \geq 0$ are slack variables.

C. CRB Transformation

It should be noted that $\text{CRB}(\theta)$ is non-convex in $\mathbf{R}_x$, due to the fractional structure coupled with quadratic forms. The constraint (20d) can be transformed by taking reciprocals of both sides, which

yields the equivalent reformulation:

$$\text{Tr}(\dot{\mathbf{A}}(\theta)\mathbf{R}_x\dot{\mathbf{A}}^H(\theta)) - \frac{\sigma_R^2}{2\rho\text{SNR}} - \frac{\|\text{Tr}(\dot{\mathbf{A}}(\theta)\mathbf{R}_x\mathbf{A}^H(\theta))\|^2}{\text{Tr}(\mathbf{A}(\theta)\mathbf{R}_x\mathbf{A}^H(\theta))} \geq 0. \quad (44)$$

By employing the Schur complement technique [16] [26], (44) can be equivalently reformulated as the following linear matrix inequality:

$$\begin{bmatrix} \text{Tr}(\dot{\mathbf{A}}(\theta)\mathbf{R}_x\dot{\mathbf{A}}^H(\theta)) - \frac{\sigma_R^2}{2\rho\text{SNR}} & \text{Tr}(\dot{\mathbf{A}}(\theta)\mathbf{R}_x\mathbf{A}^H(\theta)) \\ \text{Tr}(\dot{\mathbf{A}}(\theta)\mathbf{R}_x\mathbf{A}^H(\theta)) & \text{Tr}(\mathbf{A}(\theta)\mathbf{R}_x\mathbf{A}^H(\theta)) \end{bmatrix} \succcurlyeq 0. \quad (45)$$

To address the challenges posed by quadratic terms, $\mathbf{R}_x$ is used as the slack variable and the matrix equality constraint $\mathbf{R}_x = \mathbf{P}\mathbf{P}^H$ is relaxed into a linear matrix inequality via the Schur complement, formally expressed as:

$$\begin{bmatrix} \mathbf{R}_x & \mathbf{P} \\ \mathbf{P}^H & \mathbf{I} \end{bmatrix} \succcurlyeq 0. \quad (46)$$

D. AO algorithm based WMMSE framework

Based on the above, the original max-min optimization problem (20) can be formulated as:

$$\max_{\mathbf{P}, \mathbf{c}, R_m, \mathbf{g}, \mathbf{w}, \Delta, \mathbf{R}_x} R_m \quad (47)$$

$$\text{s. t. } \sum_{i=1}^K C_k \leq 1 - \xi_{c,k}(g_{c,k}, w_{c,k}), \quad \forall k \in \mathcal{K} \quad (47a)$$

$$R_m \leq C_k + 1 - \xi_k(g_k, w_k), \quad \forall k \in \mathcal{K} \quad (47b)$$

$$(20a), (20c), (42a), (43a), (45), (46)$$

where $\mathbf{g} \triangleq \{g_{c,k}, g_k | k \in \mathcal{K}\}$, $\mathbf{w} \triangleq \{w_{c,k}, w_k | k \in \mathcal{K}\}$ and $\Delta \triangleq \{\Delta_{c,k}, \Delta_k | k \in \mathcal{K}\}$. Analyzing problem (47), we can observe that the weights and equalizers in the WMSE constraints are decoupled, and the single optimization is enabled. The original problem is decomposed into a WMMSE subproblem and a max-min subproblem. Since minimizing the WMSE is equivalent to maximizing the rate, the optimization objectives of these two subproblems are aligned. This ensures the monotonic improvement of the objective function with each iteration, thereby enabling the solution of problem (47) via the AO principle. Specifically, $\mathbf{g}$ is optimized from (42) and (43) at first by fixed other variables. Next, substitute the obtained equalizers $\mathbf{g}$ into (25) and (26) to derive $\varepsilon_{c,k}^{MMSE}$ and ε_k^{MMSE} . Then, $\mathbf{w}$ can be obtained from $1/\varepsilon_{c,k}^{MMSE}$ and

1/ ε_k^{MMSE} . Finally, with $\mathbf{g}$ and $\mathbf{w}$ held fixed, the remaining variables are obtained by solving problem (47). As described in Algorithm 1, the procedure iteratively repeats this process until convergence.

Algorithm 1 Alternating Optimization Algorithm Based WMMSE Framework

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1: initialize:  $n \leftarrow 0$ ,  $\mathbf{P} \leftarrow \mathbf{P}^{[0]}$ ,  $R_m^{[0]} \leftarrow 0$ .
2: repeat:
3:    $n \leftarrow n + 1$ 
4:   by given  $\mathbf{P}$ , update  $g_{c,k}$ ,  $g_k$  from (42) and (43)
5:   update  $w_{c,k}$ ,  $w_k$  from  $\varepsilon_{c,k}^{MMSE}$  and  $\varepsilon_k^{MMSE}$ 
6:   solve problem (47) to obtain  $\mathbf{P}^{(n)}$ ,  $R_m^{[n]}$ ,  $\mathbf{c}^{[n]}$ ,  $\mathbf{R}_x^{[n]}$ ,  $\Delta^{[n]}$ .
7:   set  $\mathbf{P} \leftarrow \mathbf{P}^{(n)}$ 
8: until  $|R_m^{[n]} - R_m^{[n-1]}| \leq \delta_1$ 
9: output:  $\mathbf{P}^{[n]}$ ,  $\mathbf{c}^{[n]}$ ,  $R_m^{[n]}$ 

```

For the AO algorithm, initialization settings play a decisive role in the final solution performance. Determining a reasonable initial point constitutes a non-trivial challenge. To address this, a random initialization method combined with optimal value selection is adopted in this paper. Since (46) in Algorithm 1 relaxes the original equality constraints into inequality constraints, this relaxation may degrade algorithm performance. At the same time, the derivation and computation of Algorithm 1 are rather complex. To obtain a simpler and more accurate solution, we introduce a method based on SDR to circumvent these issues, while addressing channel uncertainty via the S-procedure in the next section.

V. APPROACH BASED ON S-PROCEDURE

In this section, a SDR-based approach that incorporates the S-procedure to eliminate the impact of channel uncertainty is proposed. By defining $\mathbf{P}_c = \mathbf{p}_c \mathbf{p}_c^H$, $\mathbf{P}_k = \mathbf{p}_k \mathbf{p}_k^H$, $\forall k \in \mathcal{K}$, where $\text{rank}(\mathbf{P}_c) = 1$ and $\text{rank}(\mathbf{P}_k) = 1$, the signal covariance matrix can be expressed as $\mathbf{R}_x = \mathbf{P} \mathbf{P}^H = \mathbf{P}_c + \sum_{k \in \mathcal{K}} \mathbf{P}_k$. Thus, problem (20) is equivalently transformed into an SDP problem.

To address the non-convexity of the problem and the bounded uncertainty of the channel, the problem is transformed into a tractable convex optimization form. The SCA is employed to handle the non-convex components and utilize the S-procedure to robustify against channel uncertainties. The radar CRB constraints are converted into LMI form by using the Schur complement in Section IV, thereby systematically enabling the effective solution to be derived.

A. Rate function transformation by SCA

At first, (7) and (8) can be reformulated as:

$$R_{c,k}(\mathbf{h}_k, \mathbf{P}) = \log_2(\sigma_n^2 + \text{Tr}(\mathbf{h}_k^H (\mathbf{P}_c + \sum_{j \in \mathcal{K}} \mathbf{P}_j) \mathbf{h}_k)) - \log_2(\sigma_n^2 + \text{Tr}(\mathbf{h}_k^H \sum_{j \in \mathcal{K}} \mathbf{P}_j \mathbf{h}_k)), \quad (48)$$

$$R_k(\mathbf{h}_k, \mathbf{P}) = \log_2(\sigma_n^2 + \text{Tr}(\mathbf{h}_k^H \sum_{j \in \mathcal{K}} \mathbf{P}_j \mathbf{h}_k)) - \log_2(\sigma_n^2 + \text{Tr}(\mathbf{h}_k^H \sum_{j \in \mathcal{K}, j \neq k} \mathbf{P}_j \mathbf{h}_k)). \quad (49)$$

Note the difference from a concave function, which is a non-convex function. To this end, the upper and lower bounds of SINRs are derived in (48) and (49) by using auxiliary variables a_k , b_k , c_k , d_k :

$$\sigma_n^2 + \text{Tr}(\mathbf{h}_k^H (\mathbf{P}_c + \sum_{j \in \mathcal{K}} \mathbf{P}_j) \mathbf{h}_k) \geq a_k, \quad (50)$$

$$\sigma_n^2 + \text{Tr}(\mathbf{h}_k^H \sum_{j \in \mathcal{K}} \mathbf{P}_j \mathbf{h}_k) \leq b_k, \quad (51)$$

$$\sigma_n^2 + \text{Tr}(\mathbf{h}_k^H \sum_{j \in \mathcal{K}} \mathbf{P}_j \mathbf{h}_k) \geq c_k, \quad (52)$$

$$\sigma_n^2 + \text{Tr}(\mathbf{h}_k^H \sum_{j \in \mathcal{K}, j \neq k} \mathbf{P}_j \mathbf{h}_k) \leq d_k. \quad (53)$$

Using these four bounds, the lower bound of $R_{c,k}(\mathbf{h}_k, \mathbf{P})$ and $R_k(\mathbf{h}_k, \mathbf{P})$ are derived as:

$$R_{c,k}(\mathbf{h}_k, \mathbf{P}) \geq \frac{1}{\ln 2} (\ln a_k - \ln b_k), \quad (54)$$

$$R_k(\mathbf{h}_k, \mathbf{P}) \geq \frac{1}{\ln 2} (\ln c_k - \ln d_k). \quad (55)$$

The difference of logarithmic functions in (52) and (53) is non-convex. We employ first-order Taylor series approximation via SCA to relax the constraints, which can be expressed as:

$$\bar{R}_{c,k}(a_k, b_k) = \frac{1}{\ln 2} (\ln a_k - (\ln b_k^m + \frac{b_k - b_k^m}{b_k^m})) \leq R_{c,k}(\mathbf{h}_k, \mathbf{P}), \quad (56)$$

$$\bar{R}_k(c_k, d_k) = \frac{1}{\ln 2} (\ln c_k - (\ln d_k^m + \frac{d_k - d_k^m}{d_k^m})) \leq R_k(\mathbf{h}_k, \mathbf{P}), \quad (57)$$

where m denotes the m -th iteration in the optimization process.

At this stage, we have successfully convexified the originally non-convex rate function. However, the persistent channel uncertainties in (50)-(53) remain. To resolve this, the S-procedure is introduced to eliminate these uncertainties.

B. Uncertainty elimination by S-procedure

To mitigate the impact of imperfect CSI induced by (20e), we introduce **Lemma 2** [24].

Lemma 2: Let a function

$$f_m(\mathbf{z}) = \mathbf{z}^H \mathbf{A}_m \mathbf{z} + 2\mathcal{R}\{\mathbf{b}_m^H \mathbf{z}\} + c_m, \quad m = [1, 2]$$

where $\mathbf{z} \in \mathbb{C}^{N \times 1}$, $\mathbf{A}_m \in \mathbb{H}^{N \times N}$, $\mathbf{b}_m \in \mathbb{C}^{N \times 1}$ and $c_m \in \mathbb{R}$. Then the implication $f_1(\mathbf{z}) \leq 0 \Rightarrow f_2(\mathbf{z}) \leq 0$ holds if and only if there exists λ such that

$$\lambda \begin{bmatrix} \mathbf{A}_1 & \mathbf{b}_1 \\ \mathbf{b}_1^H & c_1 \end{bmatrix} - \begin{bmatrix} \mathbf{A}_2 & \mathbf{b}_2 \\ \mathbf{b}_2^H & c_2 \end{bmatrix} \succcurlyeq 0$$

provided that there exists a vector $\hat{\mathbf{z}}$ such that $f_m(\hat{\mathbf{z}}) \leq 0$.

Based on **Lemma 2**, (50) is reformulated as follows:

$$\mathbf{h}_k^H (\mathbf{P}_c + \sum_{j \in \mathcal{K}} \mathbf{P}_j) \mathbf{h}_k \geq a_k - \sigma_n^2. \quad (58)$$

Moreover, due to the CSI uncertainty, (20e) and (58) can be reformulated as:

$$\tilde{\mathbf{h}}_k^H \mathbf{U}_m \tilde{\mathbf{h}}_k + 2\mathcal{R}\{\mathbf{u}_m^H \tilde{\mathbf{h}}_k\} + v_m \leq 0, \quad m = [1, 2] \quad (59)$$

where $\mathbf{U}_1 = \mathbf{I}_{N_t}$, $\mathbf{U}_2 = -(\mathbf{P}_c + \sum_{j \in \mathcal{K}} \mathbf{P}_j)$, $\mathbf{u}_1 = \mathbf{0}$, $\mathbf{u}_2 = -(\mathbf{P}_c + \sum_{j \in \mathcal{K}} \mathbf{P}_j) \hat{\mathbf{h}}_k$, $v_1 = -\epsilon_{e,k}^2$ and $v_2 = a_k - \hat{\mathbf{h}}_k^H (\mathbf{P}_c + \sum_{j \in \mathcal{K}} \mathbf{P}_j) \hat{\mathbf{h}}_k - 1$. According to S-procedure, (59) can be finally transformed as:

$$\begin{bmatrix} \mu_{a,k} \mathbf{U}_1 - \mathbf{U}_2 & -\mathbf{u}_2 \\ -\mathbf{u}_2^H & \mu_{a,k} v_1 - v_2 \end{bmatrix} \succcurlyeq 0, \quad (60)$$

where $\mu_{a,k} \geq 0$ are slack variables.

Similarly, (51)-(53) can be transformed into equivalent LMI forms [27] through analogous derivations

by defining the slack variables $\mu_{b,k}$, $\mu_{c,k}$ and $\mu_{d,k}$, as specified below:

$$\begin{bmatrix} \mu_{b,k} \mathbf{I}_{N_t} - \sum_{j \in \mathcal{K}} \mathbf{P}_j & - \sum_{j \in \mathcal{K}} \mathbf{P}_j \hat{\mathbf{h}}_k \\ -(\sum_{j \in \mathcal{K}} \mathbf{P}_j \hat{\mathbf{h}}_k)^H & v_{b,k} \end{bmatrix} \succcurlyeq 0, \quad (61)$$

$$\begin{bmatrix} \mu_{c,k} \mathbf{I}_{N_t} + \sum_{j \in \mathcal{K}} \mathbf{P}_j & \sum_{j \in \mathcal{K}} \mathbf{P}_j \hat{\mathbf{h}}_k \\ (\sum_{j \in \mathcal{K}} \mathbf{P}_j \hat{\mathbf{h}}_k)^H & v_{c,k} \end{bmatrix} \succcurlyeq 0, \quad (62)$$

$$\begin{bmatrix} \mu_{d,k} \mathbf{I}_{N_t} - \sum_{j \in \mathcal{K}, j \neq k} \mathbf{P}_j & - \sum_{j \in \mathcal{K}, j \neq k} \mathbf{P}_j \hat{\mathbf{h}}_k \\ -(\sum_{j \in \mathcal{K}, j \neq k} \mathbf{P}_j \hat{\mathbf{h}}_k)^H & v_{d,k} \end{bmatrix} \succcurlyeq 0, \quad (63)$$

where $v_{b,k} = -\mu_{b,k} \epsilon_{e,k}^2 - \hat{\mathbf{h}}_k^H \sum_{j \in \mathcal{K}} \mathbf{P}_j \hat{\mathbf{h}}_k - 1 + b_k$, $v_{c,k} = -\mu_{c,k} \epsilon_{e,k}^2 + \hat{\mathbf{h}}_k^H \sum_{j \in \mathcal{K}} \mathbf{P}_j \hat{\mathbf{h}}_k + 1 - c_k$, $v_{d,k} = -\mu_{d,k} \epsilon_{e,k}^2 - \hat{\mathbf{h}}_k^H \sum_{j \in \mathcal{K}, j \neq k} \mathbf{P}_j \hat{\mathbf{h}}_k - 1 + d_k$.

Based on the above, the original max-min optimization problem (20) can be rewritten as follows:

$$\max_{\mathbf{P}_c, \{\mathbf{P}_k\}_{k \in \mathcal{K}}, \mathbf{c}, R_m, \mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}, \boldsymbol{\mu}} R_m \quad (64)$$

$$\text{s. t. } \mathbf{P}_c \succcurlyeq 0, \mathbf{P}_k \succcurlyeq 0, \quad \forall k \in \mathcal{K} \quad (64a)$$

$$\text{rank}(\mathbf{P}_c) = 1, \text{rank}(\mathbf{P}_k) = 1, \quad \forall k \in \mathcal{K} \quad (64b)$$

$$\text{diag}(\mathbf{P}_c + \sum_{j \in \mathcal{K}} \mathbf{P}_j) = \frac{P_t \mathbf{1}^{N_t \times 1}}{N_t}, \quad \forall k \in \mathcal{K} \quad (64c)$$

$$\sum_{i=1}^K C_k \leq \bar{R}_{c,k}(a_k, b_k), \quad \forall k \in \mathcal{K} \quad (64d)$$

$$R_m \leq C_k + \bar{R}_k(c_k, d_k), \quad \forall k \in \mathcal{K} \quad (64e)$$

$$\boldsymbol{\mu} \geq 0 \quad (64f)$$

$$(20a), (45), (60) - (63)$$

where $\mathbf{a} \triangleq \{a_k | k \in \mathcal{K}\}$, $\mathbf{b} \triangleq \{b_k | k \in \mathcal{K}\}$, $\mathbf{c} \triangleq \{c_k | k \in \mathcal{K}\}$, $\mathbf{d} \triangleq \{d_k | k \in \mathcal{K}\}$ and $\boldsymbol{\mu} \triangleq \{\mu_{a,k}, \mu_{b,k}, \mu_{c,k}, \mu_{d,k} | k \in \mathcal{K}\}$.

Due to the rank-one constraint in (64b), the objective function in (64) remains a non-convex optimization problem. Directly dealing with the rank-one constraint is challenging, for the rank function is non-continuous and non-differentiable. To address this issue, we incorporate the constraint as a penalty term in the objective function and iteratively adjust the penalty factor until the constraint is completely satisfied [4]. The benefit is that the rank-1 constraint is not relaxed during the optimization process. Thus, the performance degradation and the inability to reconstruct the rank-1 solution are avoided.

C. Rank-one Constraint Transformation by Penalty Function

Initially, the rank-one constraint is replaced by the equivalent equation $Tr(\mathbf{P}_c) = \|\mathbf{P}_c\|$ and $Tr(\mathbf{P}_k) = \|\mathbf{P}_k\|, \forall k \in \mathcal{K}$. By adding it as the penalty to the objective function, where the penalty term takes the form:

$$\Psi_{\text{rank}} = \frac{1}{\eta} \{ [Tr(\mathbf{P}_c) - \|\mathbf{P}_c\|] + \sum_{j \in \mathcal{K}} [Tr(\mathbf{P}_j) - \|\mathbf{P}_j\|] \}, \quad (65)$$

where η represents the penalty factor. The penalty factor reflects the impact of the penalty term on the entire optimization problem. When η is small, the imposed constraints are better satisfied by the penalty term in the final solution. However, the cost is that the objective function is heavily affected by the penalty. Therefore, the strength of the penalty can be gradually increased during the optimization process. The algorithm prioritizes constraint satisfaction as the optimization advances progressively. The formula can be obtained:

$$\eta^{(n+1)} = \lambda \cdot \eta^{(n)}. \quad (66)$$

However, since the 2-norm function exists, (65) is still non-concave. To derive the lower bound and simplify the solution process, the first-order Taylor expansion is used to address the non-convexity. By taking $\mathbf{P}_c$ as an example, when $\|\mathbf{P}_c\|_{lb}$ is the lower bound of the 2-norm for $\mathbf{P}_c$ in (63), there is $\|\mathbf{P}_c\| \geq \|\mathbf{P}_c\|_{lb}$. Thus, we can get:

$$\|\mathbf{P}_c\|_{lb} = \|\mathbf{P}_c^{(n)}\| + Tr(v_{\max,c}^{(n)}(v_{\max,c}^{(n)})^H(\mathbf{P}_c - \mathbf{P}_c^{(n)})), \quad (67)$$

where $v_{\max,c}^{(n)}$ is the eigenvector at n -th optimization iteration corresponding to the largest eigenvalue of $\mathbf{P}_c^{(n)}$. Thus, (65) can be reformulated as:

$$\Psi_{\text{rank}}^{(n)} = \frac{1}{\eta} \{ [Tr(\mathbf{P}_c) - (\|\mathbf{P}_c\|_{lb})] + \sum_{j \in \mathcal{K}} [Tr(\mathbf{P}_j) - (\|\mathbf{P}_j\|_{lb})] \}. \quad (68)$$

The following optimization problem at the n -th iteration is a convex problem and is expressed as:

$$\begin{aligned} & \max_{\mathbf{P}_c, \{\mathbf{P}_k\}_{k \in \mathcal{K}}, \mathbf{c}, R_m, \mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}, \mu} R_m - \Psi_{\text{rank}}^{(n)} \\ & \text{s. t. } (20a), (45), (60) - (63), \\ & \quad (64a), (64c) - (64f) \end{aligned} \quad (69)$$

Note that $\Psi_{\text{rank}}^{(n)}$ are updated at every optimization step by solving the problem (69). The convergence of the algorithm is guaranteed by $\Psi_{\text{rank}}^{(n)}$, thereby ensuring that the final obtained solution satisfies the rank-one condition. Overall step of the proposed algorithm is described in Algorithm 2. It should be noted that the convex problem (69) can be solved through CVX toolbox [28].

Algorithm 2 MMF-rate Algorithm Based SDR And S-procedure

```

1: initialize:  $\mathbf{P}_c^{(0)}$ ,  $\{\mathbf{P}_k^{(0)}\}_{k \in \mathcal{K}}$ , set the outer layer iteration count  $m = 0$  and inner layer iteration count  $n = 0$ .
2: repeat:
3:    $m \leftarrow m + 1$ 
4:   repeat:
5:      $n \leftarrow n + 1$ 
6:     update  $\mathbf{P}_c^*$ ,  $\{\mathbf{P}_k^*\}_{k \in \mathcal{K}}$ ,  $\Psi_{\text{rank}}^*$  by solving (69) based on  $\mathbf{P}_c^{(n-1)}$ ,  $\{\mathbf{P}_k^{(n-1)}\}_{k \in \mathcal{K}}$ 
7:     update  $\mathbf{P}_c^{(n)} \leftarrow \mathbf{P}_c^*$ ,  $\{\mathbf{P}_k^{(n)}\}_{k \in \mathcal{K}} \leftarrow \{\mathbf{P}_k^*\}_{k \in \mathcal{K}}$ 
8:     until the reduction of the objective function value falls below a predefined threshold  $\delta_2$ 
9:     update  $\mathbf{P}_c^{(0)} \leftarrow \mathbf{P}_c^{(n)}$ ,  $\{\mathbf{P}_k^{(0)}\}_{k \in \mathcal{K}} \leftarrow \{\mathbf{P}_k^{(n)}\}_{k \in \mathcal{K}}$ ,  $\Psi_{\text{rank}}^{(m)} \leftarrow \Psi_{\text{rank}}^*$ 
10:    update  $\eta$  by (66)
11:  until  $|\Psi_{\text{rank}}^{(*)}| \leq \delta_3$ 
12: output:  $\mathbf{P}_c^*$ ,  $\{\mathbf{P}_k^*\}_{k \in \mathcal{K}}$ ,  $\mathbf{c}^*$ ,  $R_m^*$ ,  $\mathbf{a}^*$ ,  $\mathbf{b}^*$ ,  $\mathbf{c}^*$ ,  $\mathbf{d}^*$ ,  $\mu^*$ 

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VI. NUMERICAL RESULTS

A. Simulation Environment

In this section, the superior performance of the proposed RSMA-ISAC under imperfect CSIT is validated through the numerical simulation results. The BS employs $N_t = 6$ and $N_r = 8$ antennas arranged in a ULA structure with half-wavelength spacing, to serve $K = 4$ communication users and track one radar target, whose angles are set as 0° . The antenna power is set as $P_t = 20$ dB, the noise variances σ_n^2 and σ_R^2 is set as 0 dB [9]. The SNR of radar system is calculated by $\text{SNR}_{\text{radar}} = -20$ dB [14]. The signal frame length is set as $L = 1024$. The channel matrix $\mathbf{H}$ is modeled by combining small-scale fading $\mathbf{G}$ and large-scale fading $\mathbf{D}$, formulated as $\mathbf{H} = \mathbf{G} \cdot \mathbf{D}^{\frac{1}{2}}$. Specifically, $\mathbf{G}$ follows an i.i.d. complex Gaussian distribution $\mathbf{G} \sim \mathcal{CN}(0, 1)$ to model fast fading, while $\mathbf{D}$ represents distance-based path loss calculated as $\mathbf{D}_k(\text{dB}) = 32.6 + 36.7 \log_{10}(d_k)$, where d_k is the distance from the user to the BS (uniformly distributed between 50-150 meters). In addition, to account for imperfect CSIT, the channel uncertainty is modeled using a bounded error model, where the channel estimation error $\tilde{\mathbf{H}}$ satisfies the norm constraint $\|\tilde{\mathbf{H}}\| \leq \epsilon_e$, with ϵ_e representing the maximum allowable error magnitude.

B. Baseline

For comparison, we consider the SDMA-assited ISAC system as a baseline. Specifically, SDMA-assisted DFRC can be achieved by disabling the common stream in (1) and (2), which is equivalent to setting $\mathbf{p}_c = 0$, where the achievable rate at user k is given by:

$$R_k^b = \log_2\left(1 + \frac{|\mathbf{h}_k^H \mathbf{p}_k|^2}{\sum_{i=1, i \neq k}^K |\mathbf{h}_k^H \mathbf{p}_i|^2 + \sigma_n^2}\right), \forall k \in \mathcal{K} \quad (70)$$

The corresponding optimization problem of maximizing the fairness rate among users while maintaining the accuracy of radar echo detection can be expressed as:

$$\begin{aligned} \max_{\mathbf{P}, \mathbf{c}} \quad & \min R_k^b \\ \text{s. t.} \quad & \text{diag}(\mathbf{P}\mathbf{P}^H) = \frac{P_t \mathbf{1}^{N_t \times 1}}{N_t} \\ & \text{CRB}(\theta) \leq \rho \\ & \mathbf{h}_k \in \mathbb{H}_k \end{aligned} \quad (71)$$

This problem can be easily solved by using Algorithm 1 and Algorithm 2.

C. Convergence analyze

The convergence performance of Algorithm 1 is shown in Fig.2 with $\rho = 5e^{-3}$ and $\epsilon_e = 0.15$ for each user. It is clear that the convergence of the objective values within the proposed AO-based algorithm is guaranteed; convergence is typically achieved within 50 iterations for both RSMA and SDMA schemes.

Under the same simulation settings, Fig.3 shows the convergence performance of Algorithm 2. The weight parameter is set $\eta = 1e^4$ to achieve an optimal balance between the penalty function and the primary objective function. Furthermore, λ is set from 0.2 to the updated η from (64), so that the value of Ψ_{rank} progressively increases during iterations to strengthen the influence of the penalty term. For both the RSMA-ISAC and SDMA-ISAC schemes, all proposed algorithms exhibit rapid convergence with respect to both the main objective function and the penalty terms. Specifically, the penalty term approaches zero by the fifth iteration and converges completely within approximately ten iterations. This validates that the proposed algorithm can guarantee the creation of a rank-one solution. The results demonstrate that the algorithm obtains a suboptimal solution while effectively preserving rank-one constraints.

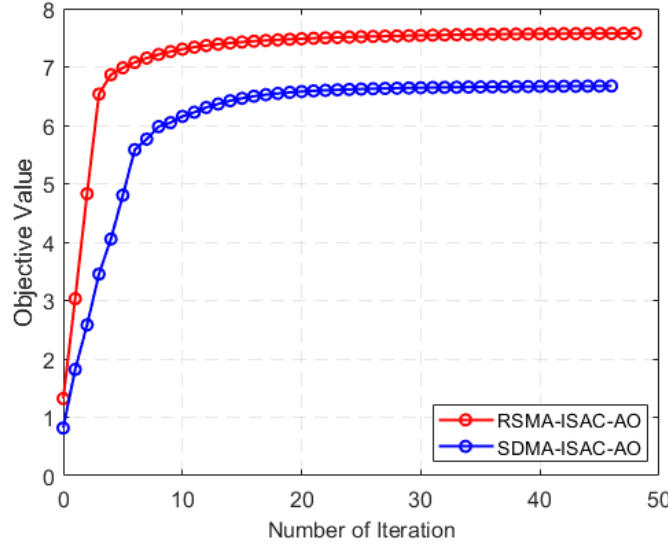


Fig. 2: Convergence performance of Algorithm 1 ($\delta_1 = 1e^{-4}$).

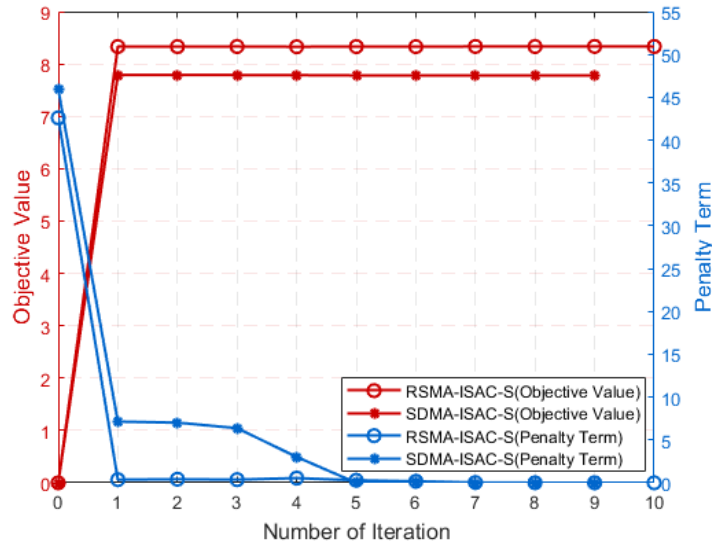


Fig. 3: Convergence of Algorithm 2 ($\delta_2 = 1e^{-4}$, $\delta_3 = 1e^{-4}$).

D. Communication performance

The critical impact of CSI on the rate performance of the system is demonstrated in Fig. 4. Specifically, Fig. 4 (a) and Fig. 4 (b) illustrate the relationship between the user's MMF-rate versus the SNR ratio under the estimation errors $\epsilon_e = 0.1$ and $\epsilon_e = 0.5$, respectively.

As observed in Fig. 4, RSMA-ISAC consistently achieves higher spectral efficiency than SDMA-ISAC, with this performance advantage progressively widening as the SNR increases. Specifically, Fig. 4 (a) illustrates that when the channel estimation error $\epsilon_e = 0.1$ and SNR reach 30 dB, the achievable MMF-rates of users in the RSMA-ISAC system are 9.69 bps/Hz and 8.54 bps/Hz, respectively, whereas those in

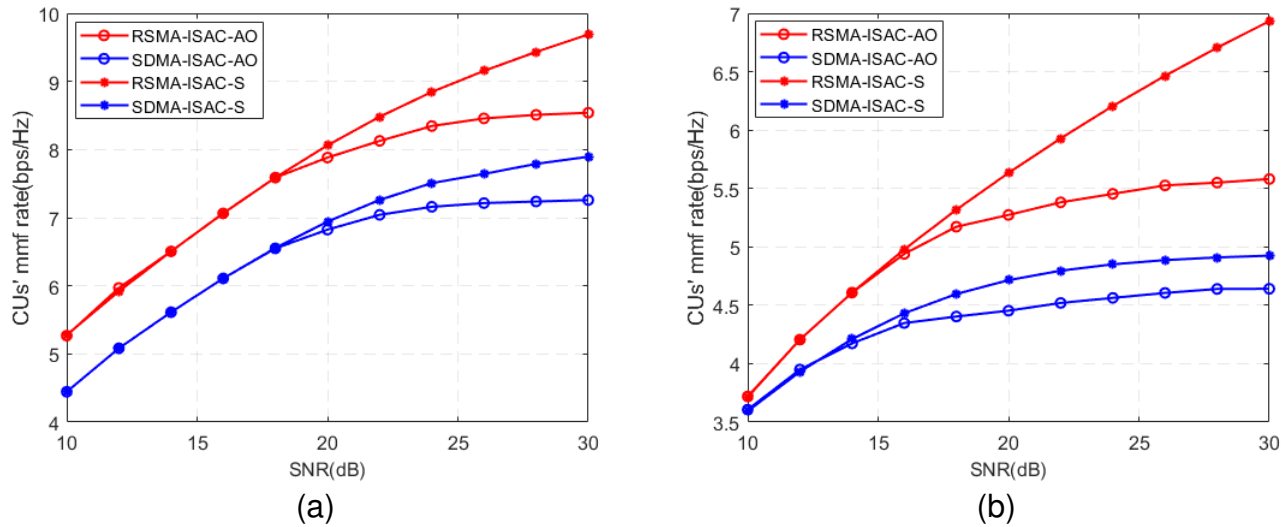


Fig. 4: Performance comparison of the two algorithms in terms of communication rate, $\text{SNR}_{\text{radar}} = -20\text{dB}$, $\rho = 5e^{-3}$. (a) $\epsilon_e = 0.1$. (b) $\epsilon_e = 0.5$.

the SDMA-ISAC system are 7.89 bps/Hz and 7.26 bps/Hz. At the SNR of 20 dB, the achievable MMF-rates of RSMA-ISAC users decrease to 8.06 bps/Hz and 7.88 bps/Hz, whereas SDMA-ISAC delivers only 6.94 bps/Hz and 6.82 bps/Hz for its users. Although both systems exhibit comparable performance in low SNR scenarios (≤ 20 dB), the MMF-rates of the users in RSMA-ISAC is about 5.27 bps/Hz, in contrast to 4.44 bps/Hz in SDMA-ISAC system.

Additionally, as channel estimation errors increase, the performance of both RSMA-ISAC and SDMA-ISAC schemes degrades significantly. However, SDMA-ISAC exhibits a more pronounced performance degradation due to channel estimation errors compared with RSMA-ISAC. As illustrated in Fig. 4(a), when the channel estimation errors $\epsilon_e = 0.1$ and the SNR is 30 dB, the achievable MMF-rates of users in the RSMA-ISAC system are 9.69 bps/Hz and 8.54 bps/Hz, respectively, whereas those in the SDMA-ISAC are 7.89 bps/Hz and 7.26 bps/Hz. When the channel estimation errors increase to 0.5, the achievable MMF-rates of RSMA-ISAC users decrease by 28.5% and 34.6%, respectively, while those of SDMA-ISAC users drop by 37.6% and 36.1%.

Furthermore, Algorithm 2 outperforms Algorithm 1 in terms of the MMF rate, especially at high SNR regime. As depicted in Fig. 4 (a), at a transmit power of 30 dB and a channel estimation error of $\epsilon_e = 0.1$, the MMF rates of the RSMA-ISAC and SDMA-ISAC achieved by Algorithm 2 are 1.15 bps/Hz and 0.64 bps/Hz higher than those of Algorithm 1, respectively. When the channel estimation error reaches 0.5, the corresponding performance obtained by Algorithm 2 in the two systems is 1.35 bps/Hz and 0.28 bps/Hz

higher than that of Algorithm 1, respectively. Notably, as the SNR increases, Algorithm 2 enables the MMF rates to increase more rapidly.

Fig. 5 shows the relationship between the MMF rate and channel estimation errors. The RSMA-ISAC framework exhibits enhanced robustness under imperfect CSIT conditions.

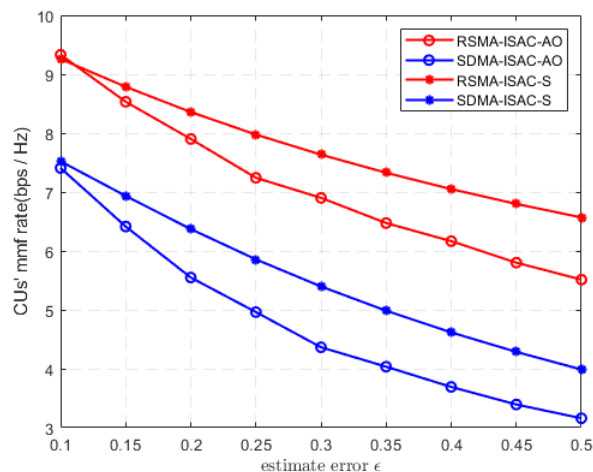


Fig. 5: MMF rate performance under different estimate error.

As illustrated in Fig. 5, the RSMA-ISAC system demonstrates significantly superior robustness to channel estimation errors than the SDMA-ISAC system. Specifically, when employing Algorithm 1, the increase in the channel estimation error from $\epsilon_e = 0.1$ to $\epsilon_e = 0.5$ results in a performance degradation of approximately 56% for the SDMA-ISAC system, whereas the RSMA-ISAC system only exhibits 32.4% reduction in MMF rate. Under the same operating conditions with Algorithm 2, the MMF rate of the SDMA-ISAC scheme decreases by about 47%, whereas that of the RSMA-ISAC scheme merely drops by 28.8%.

Having established the superiority of Algorithm 2 in high-SNR and high channel estimation error regimes, we now focus on Algorithm 2, as their performance is more representative for practical scenarios. Fig. 6 investigates the impact of the number of users on system performance.

For the sake of consistent comparative analysis, we assume that the estimation errors across all users are equal. It can be observed that the RSMA-ISAC system maintains markedly superior performance to the SDMA-ISAC system, irrespective of the number of users.

Additionally, as the SNR increases, the MMF rates of both the RSMA-ISAC and the SDMA-ISAC systems exhibit an upward trend. However, the MMF rate of the SDMA-ISAC system slows notably with increasing SNR, particularly in scenarios with a large number of users. RSMA-ISAC maintains strong

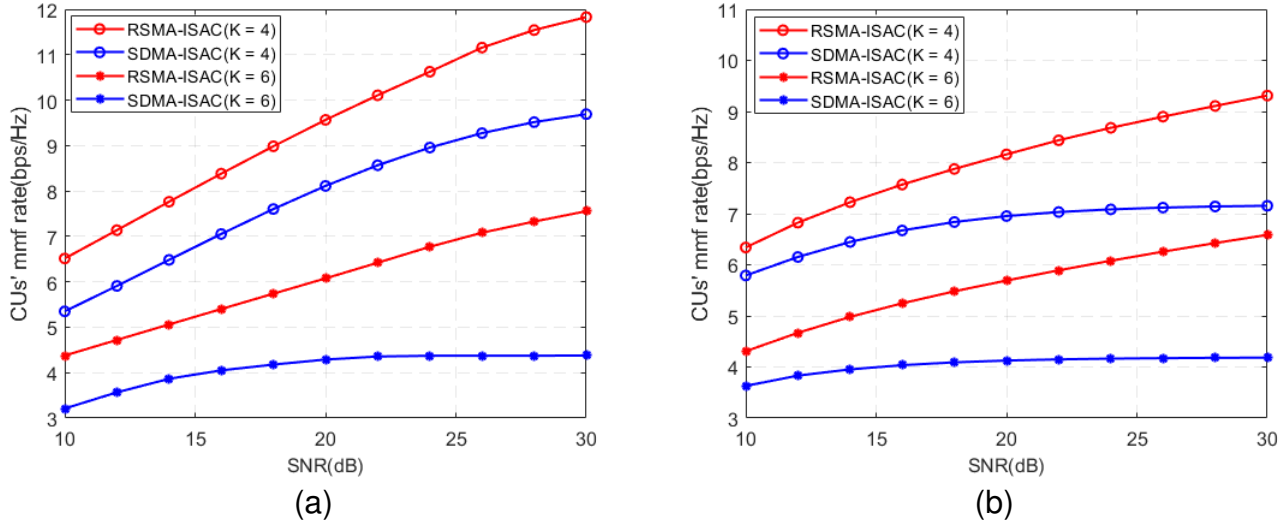


Fig. 6: Algorithm 2 rate performance for $K, N_t = 4$ and 6 , $N_r = 9$, $\text{SNR}_{\text{radar}} = -20\text{dB}$, $\rho = 5e^{-4}$. (a) $\epsilon_e = 0.1$. (b) $\epsilon_e = 0.5$.

robustness in large-scale user scenarios. As shown in Fig. 6 (a) and Fig. 6 (b), under both the moderate error condition ($\epsilon_e = 0.1$) and the severe error conditions ($\epsilon_e = 0.5$), as SNR increases, the RSMA-ISAC system achieves a significantly higher MMF rate growth rate than the SDMA-ISAC system when the number of users is set to 6.

The RSMA-ISAC system outperforms the SDMA-ISAC system in terms of robustness. As a flexible multiple-access technology, RSMA-ISAC can maintain strong robustness under severe channel uncertainty conditions. As channel conditions degrade, the MMF rate of the RSMA-ISAC system remains relatively high, whereas that of the SDMA-ISAC system drops to an extremely low level. Specifically, as shown in Fig. 6 (a), when the channel estimation error is $\epsilon_e = 0.1$, the RSMA-ISAC system achieves MMF rates of 11.83 bps/Hz and 9.69 bps/Hz for the $K = 4$ and $K = 6$ user scenarios, respectively. When the channel estimation error is $\epsilon_e = 0.5$, Fig. 6 (b) shows that the corresponding MMF rates of the RSMA-ISAC system decrease to 9.31 bps/Hz and 6.59 bps/Hz. In contrast, for the SDMA-ISAC system, Fig. 6 (a) indicates that its MMF rates are 7.56 bps/Hz and 4.38 bps/Hz under the $\epsilon_e = 0.1$ condition for $K = 4$ and $K = 6$, respectively. When the channel estimation error rises to $\epsilon_e = 0.5$, the MMF rate of SDMA-ISAC reduces to 7.15 bps/Hz and 4.18 bps/Hz, as illustrated in Fig. 6 (b).

E. Performance trade-off

The trade off between communication and sensing is illustrated in Fig. 7, i.e., the relationship between sensing estimation accuracy - CRB and communication performance - MMF rate.

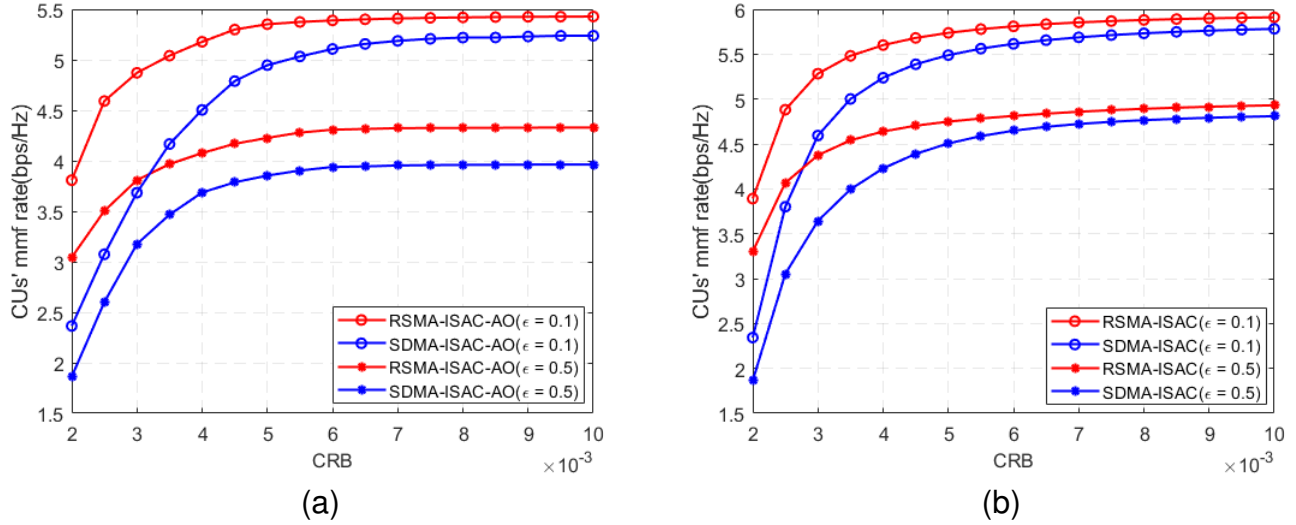


Fig. 7: Trade-off performance between communication and sensing, 20 dB transmit power and -20 dB radar SNR.(a) Algorithm 1. (b) Algorithm 2.

The RSMA-ISAC system can achieve a superior tradeoff between communication and sensing under practically imperfect CSIT conditions. As observed, RSMA-ISAC achieves a higher MMF rate than SDMA-ISAC at the left inflection point, underscoring its superior performance. Furthermore, in a general scenario, when the CRB falls below a specific threshold of approximately $5e^{-3}$, any further improvement in sensing accuracy will trigger a sharp degradation in communication performance. Specifically, as depicted in Fig. 7 (a), when the channel estimation error is $\epsilon_e = 0.1$, the MMF rate of RSMA-ISAC decreases by approximately 29%, whereas that of SDMA-ISAC drops by as much as 54%. At a higher channel estimation error of $\epsilon_e = 0.5$, the corresponding MMF rate reductions for the two systems are 31% and 56%, respectively. A consistent conclusion can be drawn from Fig. 7 (b). Under $\epsilon_e = 0.1$ conditions and CRB decreases from $5e^{-3}$ to $2e^{-3}$, the MMF rate of RSMA-ISAC experiences a reduction of roughly 27%, while the SDMA-ISAC system suffers a far more degradation of about 58%. When the channel estimation error to $\epsilon_e = 0.5$, the respective MMF rate reductions for RSMA-ISAC and SDMA-ISAC stand at 36% and 50%. These results highlight the stronger robustness of RSMA-ISAC to parameter variations.

Additionally, Algorithm 2 outperforms Algorithm 1 in terms of overall performance. Specifically, as shown in Fig. 7 (a), when the CRB is set to $10e^{-3}$, the RSMA-ISAC system achieves MMF rates of

5.43 bps/Hz and 4.32 bps/Hz with Algorithm 1 under channel estimation error conditions of $\epsilon_e = 0.1$ and $\epsilon_e = 0.5$, respectively. In Contrast, with Algorithm 2, the corresponding MMF rates of the RSMA-ISAC system increase to 5.91 bps/Hz and 4.93 bps/Hz under the same channel estimation conditions.

VII. CONCLUSIONS

This paper investigated the MMF optimization problem in RSMA-ISAC systems. The results demonstrated that RSMA-ISAC exhibits superior performance under imperfect CSIT conditions. To tackle the combined difficulties of an infinite error set and intrinsic non-convexity, we introduce the AO algorithm based on the WMMSE framework and a double-layer penalty-based SDR algorithm, respectively. The simulation demonstrated that gratified performance can be obtained when RSMA is used as the interference management strategy in ISAC. High-quality communication and sensing services can be provided simultaneously in the ISAC system. In addition, better QoS and superior fairness can be achieved. Finally, we confirmed that the proposed RSMA-ISAC is robust to imperfect CSIT. As a result, RSMA ensures that both strong and weak users receive adequate support and that overall communication and sensing performance is enhanced.

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